Characterizing Datasets: Hardness, Learning, and More

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Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

1 Core Idea: Datasets and Basic In-Distribution Testing

Ideally, we'd like to take the useful tools of Fourier Analysis (which assumes a product space) and generalize them to any distribution where the underlying super-space is a product space[1]. Though, we do not have a formal method of reasoning about this, we do not think that it is quite possible.

Rather, we will attempt to think about analysis as over a *dataset:* i.e. we will think of our distribution as being defined by a finite-sized dataset, $\mathcal{D}$, and the probability vector will be defined as

$$p(x) = \begin{cases} \frac{1}{|\mathcal{D}|} & \text{if } \ x \in \mathcal{D} \\ 0 \text{ otherwise} \end{cases} .$$

More formally, we will be over a space $\mathcal{T}^n$ (think of $\mathcal{T}$ as either $\mathbb{R}$ or the space of tokens etc.) and $\mathcal{D} \subset \mathcal{T}^n$. Then, we will be focusing on functions $f \in \mathcal{T}^n \to \mathbb{R}$. f can either be a trained model, ideal labeling function on the dataset, or some other efficiently computable function.

Then, we want to reason about Fourier coefficients as

$$\hat{f}(S) = \mathbb{E}_{x \sim \mathcal{D}}[f(x)\chi_S^{-1}(x)].$$

Further, we will abuse notation to denote $\mathcal{D} : \mathcal{T}^n \to \{0, 1\}$ as the indicator function for the inclusion within the dataset (i.e. an in-distribution tester). We will write $\hat{f}_{\mathrm{OG}}(S)$ to denote the normal ("original") Fourier coefficient:

$$\hat{f}_{\mathrm{OG}}(S) = \mathbb{E}_{x \sim \mathcal{T}^n}[f(x)\chi_S^{-1}(x)].$$

Importantly, notice that

$$\hat{f}(S) = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} f(x)\chi_S(x) = \frac{|\mathcal{T}^n|}{|\mathcal{D}|} \cdot \hat{f}_{\mathrm{OG}}(\mathcal{D} \quad \circ \quad f)$$

[1]This includes, but is not limited too, language datasets over tokens, images over pixels, and other similar data modalities.

where $\circ$ is the element wise composition. TODO: I think we need to use χ^{-1}?

$\frac{\mathcal{T}^n}{|\mathcal{D}|}$ is a normalizing constant which we will frequently arise. As such, we will denote $\frac{|\mathcal{T}|^n}{|\mathcal{D}|}$ as $C_{\mathcal{D}}$ and its inverse as $C_{\mathcal{D}}^{-1}$ for ease of notation.

We conveniently have our first lemma.

Lemma 1.1: For all $x \in \mathcal{D}$,

$$f(x) = C_{\mathcal{D}}^{-1} \sum_S \hat{f}(S)\chi_S(x)$$

Proof: First, note that $f(x) = \mathcal{D}(x) \cdot f(x)$ for $x \in \mathcal{D}$ and then $\widehat{\mathcal{D} \circ f}_{\mathrm{OG}}(S) = C_{\mathcal{D}}^{-1} \cdot \hat{f}(S)$ and as such, using the standard Fourier identity

$$\mathcal{D}(x) \cdot f(x) = \sum_S \widehat{\mathcal{D} \circ f}(S)\chi_S(x) = C_{\mathcal{D}}^{-1} \sum_S \hat{f}(S).$$

■

1.1 In Distribution Testing and Related Notions

Unfortunately, we were not able to find a good way to define property testing, *soley* over the distribution. Rather, we will need to test both in-distribution and out-of-distribution samples. Still, we will only need to test out-of-distribution samples in a very limited way: samples which are only hamming distance 1 away from in-distribution samples for most of our properties.

To denote the need for in-distribution testing, we will append $\mathcal{D}$ as a subscript to various operators.

Definition 1.1 *(Distribution-Testing Coordinate Averaging)*: Let $E_{\mathcal{D}}^i$ for $i \in n$ be the i-th in distribution operator for $x \in \mathcal{D}$:

$$E_{\mathcal{D}}^i[f](x) = \mathbb{E}_{a \in \mathcal{T}}[\mathcal{D}(x_i \mapsto a) \circ f(x^{x_i \mapsto a})].$$

Note that Definition 1.1 *zeros out* any coordinate setting which does not remain within the dataset. Intuitively, we can think about $E_{\mathcal{D}}^i$ as the generic coordinate averaging operator for a function which will test whether an input is in the dataset and output 0 if it is not.

We can now define influence:

Definition 1.2 *(i-th Coordinate Distribution-Testing Influence Operator)*:

$$\mathrm{Inf}_{\mathcal{D}}^i f = \mathbb{E}_{x \in \mathcal{D}}\left[\left(f(x) - E_{\mathcal{D}}^i f(x)\right)^2\right].$$

Proposition 1.1: We now prove that basic identities still hold as in O'Donnell's Analysis of Boolean Functions [1]:

$$\mathrm{Inf}_{\mathcal{D}}^i f = C_{\mathcal{D}} \langle \mathcal{D} \circ f, \mathcal{D} \circ (f - E_D^i) \rangle,$$

$$E_D^{i_{\mathcal{D}}^i} f = C_{\mathcal{D}} \sum_{S, S_i = 0} \hat{f}(S)\chi_S(x)$$

$$\mathrm{Inf}_{\mathcal{D}}^i f = C_{\mathcal{D}} \sum_{S, S_i \neq 0} \hat{f}(S)^2.$$

Proof: We start with the second equality. TODO: check constants Note that,

$$
\begin{aligned}
E_{\mathcal{D}}^{i_{\mathcal{D}}^{i}}(x) &= \mathbb{E}_{a\in\mathcal{T}}[\mathcal{D}(x_i \mapsto a) \quad \circ \quad f(x^{x_i\mapsto a})] \\
&= \sum_{S, S_i = 0} \hat{f}_{\mathrm{OG}}(f)(S)\chi_S(x) \\
&= C_{\mathcal{D}} \sum_{S, S_i=0} \hat{f}(S)\chi_S(x).
\end{aligned}
$$

The first equality holds as we note that,

$$
\begin{aligned}
\mathrm{Inf}^i_{\mathcal{D}} f &= C_{\mathcal{D}}^{-1}\langle f(x) - E^i_{\mathcal{D}}(x), f(x) - E^i_{\mathcal{D}}(x)\rangle \\
&= C_{\mathcal{D}}^{-1}\left(\mathbb{E}[f(x)^2] - 2E_x\left[f(x)E^i_{\mathcal{D}}(x)\right] + \mathbb{E}_{\mathcal{D}}\left[E^i_{\mathcal{D}}(x)^2\right]\right).
\end{aligned}
$$

TODO: norm constant Then, note that by the second equality and Parseval's theorem, $E_{x\sim\mathcal{T}^n}\left[f(x)E^i_{\mathcal{D}}(x)\right] = \sum_{S,S_i=0} \hat{f}_{\mathrm{OG}}(S)^2$ and $\mathbb{E}_{\mathcal{D}}\left[E^i_{\mathcal{D}}(x)^2\right] = \sum_{S,S_i=0} \hat{f}(S)^2$ as desired.

Finally, we note that as $\mathrm{Inf}^i_{\mathcal{D}} f = \mathbb{E}_{\mathcal{D}}[f(x)^2] - \mathbb{E}_{\mathcal{D}}\left[E^i_{\mathcal{D}}(x)^2\right]$, we can see that

$$
\mathbb{E}_{\mathcal{D}}[f(x)^2] - E_x\left[f(x)E^i_{\mathcal{D}}(x)\right] = \sum_S \hat{f}(S)^2 - \sum_{S,S_i=0} \hat{f}(S)^2 = \sum_{S,S_i\neq 0} \hat{f}(S)^2.
$$

■

Just as in TODO: cite , we can define total influence and get some convenient corollaries:

Definition 1.3 *(Total Influence, $\boldsymbol{I}_{\mathcal{D}}$)*:

$$
\boldsymbol{I}_{\mathcal{D}}[f] = \sum_{i\in[n]} \mathrm{Inf}^i_{\mathcal{D}}[f].
$$

We immediately get:

Proposition 1.2:

$$
\boldsymbol{I}_{\mathcal{D}}[f] = \sum_S \#S \quad \hat{f}(S)^2
$$

which, for finite groups, we get

$$
\boldsymbol{I}_{\mathcal{D}}[f] = \sum_S k \cdot W^k[f].
$$

as in page 213 of TODO: a where $\#S = |\mathrm{supp}\ (S)| = \mathrm{supp}\ (S) = \{i : S_i \neq 0\}$.

Finally, we introduce one more definition which will capture the closeness of two functions, f, g over $\mathcal{D}$.

Definition 1.4 *(In-Distribution Closeness, $\epsilon_{\mathcal{D}}$-closeness)*: We say that a function g is $\epsilon_{\mathcal{D}}$ close to f if:

$$
\mathop{\mathbb{E}}_{x\in\mathcal{D}}[(f(x) - g(x))^2] \leq \epsilon
$$

or equivalently,

$$
C_{\mathcal{D}}^{-1} \mathop{\mathbb{E}}_{x\in\mathcal{T}^{\otimes n}}[(\mathcal{D} \quad \circ \quad f(x) - \mathcal{D} \quad \circ \quad g(x))^2] \leq \epsilon.
$$

1.2 Immediate Consequences

Learning from Random Examples

We can already adopt learning low-degree functions from random examples to our setting. Specifically, if the in-distribution Fourier mass is concentrated on low-degree terms, we can learn the function from random examples drawn from $\mathcal{D}$.

TODO: this is auto-gened, check over and make right! CLAUDE CHECK

Theorem 1.1: Let $f : \mathcal{T}^n \to \mathbb{R}$ be a function such that $\sum_{S:\ |S|\ > d} \hat{f}(S)^2 \le \frac{\epsilon^2}{4}$. Then, there exists an algorithm which, given $O\left(\frac{n^d}{\epsilon^2} \log(n)\right)$ random examples from $\mathcal{D}$, outputs a function g which is $\epsilon_{\mathcal{D}}$ close to f with probability at least $\frac{2}{3}$.

One-Way Property Testing for Computable via Decision Tree

1.3 In Distribution Testing Definitions

Surprisingly, if want to characterize the complexity (or sensitivity or one of many Fourier properties) of our function f over distribution $\mathcal{D}$ and in-distribution testing, we can more or less use standard analysis.

In more detail, we will model our modified function $g : |\mathcal{T}|^n \to \mathbb{R}^{+^m} \cup \{0\}$ as

$$g(x) = \begin{cases} f(x) & \text{if } \ x \in \mathcal{D} \\ 0 & \text{otherwise} \end{cases}$$

And, taking the standard inner-product (TODO: we are over sphere/ real numbers, first take the inner-product then the other thing!)

Note that g is required to differentiate between in and out-of distribution.

Influence and Related Operators

We would like coordinate wise influence to be defined in the standard way, but adopted to our setting:

$$\operatorname{Inf}^i_{\mathcal{D}}[g] = \mathop{\mathbb{E}}_{x \in \mathcal{D}}\left[\left(g - E^i_{\mathcal{D}} g\right)^2\right]$$

where $E^i_{\mathcal{D}}$ is the i-th coordinate expectation operator:

$$\left[E^i_{\mathcal{D}} g\right](x) = \mathop{\mathbb{E}}_{x_i \in \mathcal{T}}\left[g\left(x^{i \mapsto x_i}\right)\right].$$

In words, the i-th expectation operator "averages out" the i-th coordinate over the dataset while the Influence measures the difference.

TODO: define distribution coeffs! TODO: This prob dist thing doesn't work??? Just have a subset thingy for now!!! Then, we have the following proposition:

Proposition 1.3:

$$E^i_{\mathcal{D}} g = \sum_{s : s_i \neq 0} \widehat{p_{\text{scale}} \cdot}\, f(s) \hat{f}_{\mathcal{D}}$$

Lemma 1.2: We can re-write the influence as an inner-product

$$\operatorname{Inf}^i_{\mathcal{D}}[f] = \left\langle p \cdot \left(g - E^i_{\mathcal{D}} g\right), g \right\rangle = \left\langle p \cdot \left(g - E^i_{\mathcal{D}} g\right), g - E^i_{\mathcal{D}} g \right\rangle$$

Proof: The second equality follows directly from the definition. The first is because

$$\left\langle p \cdot \left(g - E^i_{\mathcal{D}} g\right), g - E^i_{\mathcal{D}} g \right\rangle = \mathbb{E}_x\left[g^2\right] - 2\mathbb{E}_x\left[g \cdot E^i_{\mathcal{D}} f\right] + \mathbb{E}_x\left[\left(E^i_{\mathcal{D}} f\right)^2\right]$$

■

TODO: What happens if you take $f = \mathcal{D}$? TODO: Is there a monotonic question of degree over dataset vs degree over full space? (you can then know if a model class can learn it or not) (!!!!

This is a good idea. People would care a lot more !!!!) Look into: Statistical Learning theory: *distribution agnostic learning theory*

2 Orthonormal Basis Analysis

We develop a Fourier-analytic framework over general probability spaces equipped with an orthonormal basis. This generalizes both Boolean Fourier analysis (Haar basis) and Gaussian space analysis (Hermite polynomials).

2.1 General Framework

Definition 2.1 *(Probability Space with Orthonormal Basis)*: Let Ω be a measurable space with probability measure μ, and let I be an index set. An *orthonormal basis* for $L^2(\Omega, \mu)$ is a collection $\{\varphi_\alpha\}_{\alpha \in I}$ of functions $\varphi_\alpha : \Omega \to \mathbb{R}$ satisfying:

1. **Orthonormality:** $\langle \varphi_\alpha, \varphi_\beta \rangle_\mu = \mathbb{E}_{x \sim \mu}[\varphi_\alpha(x) \varphi_\beta(x)] = \delta_{\alpha\beta}$.
2. **Completeness:** Every $f \in L^2(\Omega, \mu)$ can be written as $f = \sum_{\alpha \in I} \hat{f}(\alpha) \varphi_\alpha$.

Definition 2.2 *(Fourier Coefficients)*: For $f \in L^2(\Omega, \mu)$, the *Fourier coefficient* at index α is

$$\hat{f}(\alpha) = \langle f, \varphi_\alpha \rangle_\mu = \mathbb{E}_{x \sim \mu}[f(x) \varphi_\alpha(x)]. \tag{1}$$

Theorem 2.1 (Fourier Expansion): Every $f \in L^2(\Omega, \mu)$ has a unique expansion

$$f = \sum_{\alpha \in I} \hat{f}(\alpha) \varphi_\alpha \tag{2}$$

with convergence in L^2.

Lemma 2.1 *(Parseval's Identity)*: For any $f \in L^2(\Omega, \mu)$,

$$\|f\|_2^2 = \mathbb{E}_{x \sim \mu}[f(x)^2] = \sum_{\alpha \in I} \hat{f}(\alpha)^2. \tag{3}$$

Proof: Expanding f in the orthonormal basis and using bilinearity of the inner product:

$$\begin{aligned}
\mathbb{E}_{x \sim \mu}[f(x)^2] &= \langle f, f \rangle_\mu \\
&= \left\langle \sum_\alpha \hat{f}(\alpha) \varphi_\alpha, \sum_\beta \hat{f}(\beta) \varphi_\beta \right\rangle_\mu \\
&= \sum_\alpha \sum_\beta \hat{f}(\alpha) \hat{f}(\beta) \langle \varphi_\alpha, \varphi_\beta \rangle_\mu \\
&= \sum_\alpha \sum_\beta \hat{f}(\alpha) \hat{f}(\beta) \delta_{\alpha\beta} \\
&= \sum_\alpha \hat{f}(\alpha)^2.
\end{aligned} \tag{4}$$

The interchange of sum and integral is justified by L^2 convergence. ∎

Corollary 2.1 *(Parseval for Differences)*: For $f, g \in L^2(\Omega, \mu)$,

$$\mathbb{E}_{x \sim \mu}[(f(x) - g(x))^2] = \sum_{\alpha \in I} \left(\hat{f}(\alpha) - \hat{g}(\alpha)\right)^2. \tag{5}$$

Proof: Define $h = f - g$. By linearity of the Fourier transform, $\hat{h}(\alpha) = \hat{f}(\alpha) - \hat{g}(\alpha)$. Applying Parseval's identity to h:

$$\mathbb{E}_{x \sim \mu}[(f(x) - g(x))^2] = \|h\|_2^2 = \sum_{\alpha} \hat{h}(\alpha)^2 = \sum_{\alpha} \left(\hat{f}(\alpha) - \hat{g}(\alpha)\right)^2. \tag{6}$$

■

2.2 Sensitivity

For product spaces $\Omega = \Omega_1 \times \cdots \times \Omega_n$ with product measure $\mu = \mu_1 \otimes \cdots \times \mu_n$, we can define coordinate-wise sensitivity.

Definition 2.3 *(Coordinate Averaging)*: For $x \in \Omega^n$ and coordinate $i \in [n]$, define

$$E^i[f](x) = \mathbb{E}_{a \sim \mu_i}[f(x^{i \leftarrow a})] \tag{7}$$

where $x^{i \leftarrow a}$ denotes x with the i-th coordinate replaced by a.

Definition 2.4 *(Coordinate Sensitivity)*: The *sensitivity of coordinate i* on $f \in L^2(\Omega^n, \mu)$ is

$$\mathrm{Sens}_i[f] = \mathbb{E}_{x \sim \mu}\left[\left(f(x) - E^i[f](x)\right)^2\right]. \tag{8}$$

Definition 2.5 *(Total Sensitivity)*: The *total sensitivity* of f is

$$\boldsymbol{S}[f] = \sum_{i=1}^{n} \mathrm{Sens}_i[f]. \tag{9}$$

In the literature, sensitivity is often called *influence*. We use "sensitivity" to emphasize its interpretation as measuring how much f depends on each coordinate.

2.3 Examples of Orthonormal Bases

2.3.1 Haar Basis (Boolean Cube)

Definition 2.6 *(Boolean Cube)*: The Boolean cube is $\Omega = \{-1, 1\}^n$ with the uniform measure $\mu(x) = 2^{-n}$ for all x.

Definition 2.7 *(Haar Basis / Parity Functions)*: For $S \subseteq [n]$, define the *parity function*

$$\chi_S(x) = \prod_{i \in S} x_i. \tag{10}$$

The collection $\{\chi_S\}_{S \subseteq [n]}$ forms an orthonormal basis for $L^2(\{-1, 1\}^n, \mu)$.

Proposition 2.1 *(Haar Orthonormality)*: $\langle \chi_S, \chi_T \rangle = \delta_{ST}$.

Proof: We have $\chi_S(x)\chi_T(x) = \chi_{S \Delta T}(x)$ where $S \Delta T$ is the symmetric difference. For $S \neq T$, there exists $i \in S \Delta T$, so $\mathbb{E}[\chi_{S\Delta T}] = \mathbb{E}[x_i] \cdot \mathbb{E}\left[\chi_{S \Delta T \setminus \{i\}}\right] = 0$. For $S = T$, $\chi_{S \Delta T} = \chi_{\emptyset} = 1$. ■

Proposition 2.2 *(Boolean Sensitivity)*: For $f : \{-1, 1\}^n \to \mathbb{R}$,

$$\mathrm{Sens}_i[f] = \sum_{S : i \in S} \hat{f}(S)^2. \tag{11}$$

2.3.2 Probabilist's Hermite (Gaussian Space)

Definition 2.8 *(Standard Gaussian Measure)*: The *standard Gaussian measure* gam on $\mathbb{R}$ has density

$$d\gamma(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx. \tag{12}$$

The n-dimensional Gaussian measure γ_n on $\mathbb{R}^n$ is the product measure $\gamma^{\otimes n}$.

Definition 2.9 *(Univariate Hermite Polynomials)*: The *probabilist's Hermite polynomials* $h_j : \mathbb{R} \to \mathbb{R}$ are the orthonormalized polynomials with respect to gam. The first few are:

$$h_0(x) = 1, \quad h_1(x) = x, \quad h_2(x) = \frac{x^2 - 1}{\sqrt{2}}, \quad h_3(x) = \frac{x^3 - 3x}{\sqrt{6}}. \tag{13}$$

Proposition 2.3 *(Hermite Orthonormality)*: $\langle h_j, h_k \rangle_\gamma = \delta_{jk}$.

Definition 2.10 *(Multivariate Hermite Polynomials)*: For a multi-index $\alpha \in \mathbb{N}^n$, define

$$H_\alpha(x) = \prod_{i=1}^{n} h_{\alpha_i}(x_i). \tag{14}$$

The *degree* of H_α is $|\alpha| = \sum_{i=1}^{n} \alpha_i$.

Theorem 2.2 (Hermite Expansion): Every $f \in L^2(\mathbb{R}^n, \gamma_n)$ has a unique expansion

$$f = \sum_{\alpha \in \mathbb{N}^n} \hat{f}(\alpha) H_\alpha \tag{15}$$

where $\hat{f}(\alpha) = \langle f, H_\alpha \rangle_{\gamma_n} = \mathbb{E}_{x \sim \gamma_n}[f(x) H_\alpha(x)]$.

Proposition 2.4 *(Gaussian Sensitivity via Derivatives)*: For $f \in L^2(\mathbb{R}^n, \gamma_n)$ with weak derivative $\partial_i f$,

$$\mathrm{Sens}_i[f] = \mathbb{E}_{x \sim \gamma_n}\left[(\partial_i f(x))^2\right] = \sum_{\alpha : \alpha_i \geq 1} \alpha_i \cdot \hat{f}(\alpha)^2. \tag{16}$$

Proof: We have $\partial_i H_\alpha = \sqrt{\alpha_i} H_{\alpha - e_i}$ where e_i is the i-th standard basis vector (the term vanishes if $\alpha_i = 0$). By Parseval,

$$\mathrm{Sens}_i[f] = \mathbb{E}\left[(\partial_i f)^2\right] = \sum_{\alpha : \alpha_i \geq 1} \alpha_i \hat{f}(\alpha)^2. \tag{17}$$

■

Noise Operator

Definition 2.11 *(ρ-Correlated Gaussians)*: For $\rho \in [-1, 1]$, we say (x, y) are *ρ-correlated Gaussians* if $x \sim \gamma_n$ and

$$y = \rho x + \sqrt{1 - \rho^2} z \tag{18}$$

where $z \sim \gamma_n$ is independent of x. We write $y \sim N_\rho(x)$ for the conditional distribution of y given x.

Definition 2.12 *(Ornstein-Uhlenbeck Operator)*: The *noise operator* U_ρ is defined by

$$U_\rho f(x) = \mathbb{E}_{y \sim N_\rho(x)}[f(y)]. \tag{19}$$

Proposition 2.5 *(Hermite Eigenfunction Property)*: The Hermite polynomials are eigenfunctions of U_ρ:

$$U_\rho H_\alpha = \rho^{|\alpha|} H_\alpha. \tag{20}$$

Proof: By independence of coordinates and linearity, it suffices to check the univariate case. For $y = \rho x + \sqrt{1-\rho^2} z$ with $z \sim \gamma$ independent, we have $\mathbb{E}_z\left[h_j\left(\rho x + \sqrt{1-\rho^2} z\right)\right] = \rho^j h_j(x)$ by the generating function of Hermite polynomials. ■

Invariance Principle

Theorem 2.3 (Gaussian Invariance Principle (Informal)): Let $f : \{-1,1\}^n \to \mathbb{R}$ be a multilinear polynomial with small sensitivities. Let $g : \mathbb{R}^n \to \mathbb{R}$ be its *Gaussian version*: the function with the same multilinear coefficients, viewed as Hermite coefficients. Then $f(x)$ for $x \in \{-1,1\}^n$ uniform and $g(z)$ for $z \sim \gamma_n$ have approximately the same distribution.

This principle allows us to transfer results between Boolean and Gaussian settings.

2.4 Dataset-Specific Analysis

We now adapt the above to the setting where we have a finite dataset $\mathcal{D} \subset \Omega^n$.

Definition 2.13 *(Dataset Distribution)*: Let $\mathcal{D} \subset \Omega^n$ be a finite dataset. The uniform distribution over $\mathcal{D}$ is

$$p(x) = \begin{cases} \frac{1}{|\mathcal{D}|} & \text{if } x \in \mathcal{D} \\ 0 & \text{otherwise} \end{cases}. \tag{21}$$

Definition 2.14 *(Dataset Fourier Coefficient)*: The *dataset Fourier coefficient* is

$$\hat{f}_{\mathcal{D}}(\alpha) = \mathbb{E}_{x \sim \mathcal{D}}[f(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} f(x)\varphi_\alpha(x). \tag{22}$$

Definition 2.15 *(Dataset Coordinate Averaging)*: For $x \in \mathcal{D}$, define

$$E^i_{\mathcal{D}}[f](x) = \mathbb{E}_{a \sim \mu_i}\left[\mathcal{D}\left(x^{i \leftarrow a}\right) \cdot f\left(x^{i \leftarrow a}\right)\right] \tag{23}$$

where $\mathcal{D}(y) = 1$ if $y \in \mathcal{D}$ and 0 otherwise. This operator *zeros out* any coordinate setting that leaves the dataset.

Definition 2.16 *(Dataset Sensitivity)*:

$$\mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{x \sim \mathcal{D}}\left[\left(f(x) - E^i_{\mathcal{D}}[f](x)\right)^2\right]. \tag{24}$$

Proposition 2.6 *(Sensitivity via Fourier Coefficients)*: For orthonormal bases with product structure indexed by $(\alpha_1, ..., \alpha_n)$,

$$\mathrm{Sens}^i_{\mathcal{D}}[f] = \sum_{\alpha : \alpha_i \neq 0} \hat{f}_{\mathcal{D}}(\alpha)^2. \tag{25}$$

Proof: We first show that $\mathbb{E}_{\mathcal{D}}\left[f \cdot E^i_{\mathcal{D}} f\right] = \mathbb{E}_{\mathcal{D}}\left[\left(E^i_{\mathcal{D}} f\right)^2\right]$. This follows because $E^i_{\mathcal{D}}$ is idempotent: applying it twice gives the same result as applying it once, since averaging over x_i on a function already independent of x_i has no effect.

Now expand the definition:

$$\begin{aligned}\mathrm{Sens}^i_{\mathcal{D}}[f] &= \mathbb{E}_{\mathcal{D}}\left[\left(f - E^i_{\mathcal{D}} f\right)^2\right] \\ &= \mathbb{E}_{\mathcal{D}}[f^2] - 2\mathbb{E}_{\mathcal{D}}\left[f \cdot E^i_{\mathcal{D}} f\right] + \mathbb{E}_{\mathcal{D}}\left[\left(E^i_{\mathcal{D}} f\right)^2\right] \\ &= \mathbb{E}_{\mathcal{D}}[f^2] - \mathbb{E}_{\mathcal{D}}\left[\left(E^i_{\mathcal{D}} f\right)^2\right].\end{aligned} \tag{26}$$

The Fourier expansion of $E^i_{\mathcal{D}} f$ only includes terms with $\alpha_i = 0$. By Parseval: $\mathbb{E}_{\mathcal{D}}[f^2] = \sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)^2$ and $\mathbb{E}_{\mathcal{D}}\left[\left(E^i_{\mathcal{D}} f\right)^2\right] = \sum_{\alpha:\alpha_i=0} \hat{f}_{\mathcal{D}}(\alpha)^2$.

Therefore:

$$\mathrm{Sens}^i_{\mathcal{D}}[f] = \sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)^2 - \sum_{\alpha:\alpha_i=0} \hat{f}_{\mathcal{D}}(\alpha)^2 = \sum_{\alpha:\alpha_i \neq 0} \hat{f}_{\mathcal{D}}(\alpha)^2. \tag{27}$$

■

Definition 2.17 *(Total Dataset Sensitivity)*:

$$\boldsymbol{S}_{\mathcal{D}}[f] = \sum_{i=1}^n \mathrm{Sens}^i_{\mathcal{D}}[f]. \tag{28}$$

Definition 2.18 *(Fourier Weight at Level k)*:

$$W^k_{\mathcal{D}}[f] = \sum_{|\alpha| = k} \hat{f}_{\mathcal{D}}(\alpha)^2. \tag{29}$$

Proposition 2.7 *(Total Sensitivity via Fourier Spectrum)*:

$$\boldsymbol{S}_{\mathcal{D}}[f] = \sum_\alpha |\alpha| \cdot \hat{f}_{\mathcal{D}}(\alpha)^2 = \sum_{k \geq 1} k \cdot W^k_{\mathcal{D}}[f]. \tag{30}$$

Proof: Summing the sensitivity formula over all $i \in [n]$:

$$\boldsymbol{S}_{\mathcal{D}}[f] = \sum_{i=1}^n \mathrm{Sens}^i_{\mathcal{D}}[f] = \sum_{i=1}^n \sum_{\alpha:\alpha_i \neq 0} \hat{f}_{\mathcal{D}}(\alpha)^2. \tag{31}$$

Each α with $|\alpha| = k$ appears in exactly k of the inner sums (once for each i where $\alpha_i \neq 0$). Thus $\boldsymbol{S}_{\mathcal{D}}[f] = \sum_\alpha |\alpha| \cdot \hat{f}_{\mathcal{D}}(\alpha)^2 = \sum_{k \geq 1} k \cdot W^k_{\mathcal{D}}[f]$. ■

Definition 2.19 *(Dataset Closeness)*: We say g is *ϵ-close* to f over $\mathcal{D}$ if

$$\mathbb{E}_{x \sim \mathcal{D}}\left[(f(x) - g(x))^2\right] \leq \epsilon. \tag{32}$$

Lemma 2.2 *(Parseval for Dataset Closeness)*:

$$\mathbb{E}_{x \sim \mathcal{D}}\left[(f(x) - g(x))^2\right] = \sum_\alpha \left(\hat{f}_{\mathcal{D}}(\alpha) - \hat{g}_{\mathcal{D}}(\alpha)\right)^2. \tag{33}$$

Proof: Define $h = f - g$. By linearity, $\hat{h}_{\mathcal{D}}(\alpha) = \hat{f}_{\mathcal{D}}(\alpha) - \hat{g}_{\mathcal{D}}(\alpha)$. Applying Parseval's identity over the dataset distribution:

$$\mathbb{E}_{x \sim \mathcal{D}}[h(x)^2] = \sum_\alpha \hat{h}_{\mathcal{D}}(\alpha)^2 = \sum_\alpha \left(\hat{f}_{\mathcal{D}}(\alpha) - \hat{g}_{\mathcal{D}}(\alpha)\right)^2. \tag{34}$$

■

This lemma reduces proving ϵ-closeness to bounding the sum of squared coefficient differences.

2.5 Learning Low-Degree Functions

Theorem 2.4 (Learning Low-Degree Fourier Functions): Let $f : \Omega^n \to \mathbb{R}$ be bounded with total sensitivity $\boldsymbol{S}_{\mathcal{D}}[f] \leq S$. Given $d \geq 4\frac{S}{\epsilon^2}$ and $m = O\left(\frac{n^d}{\epsilon^2} \cdot \log n\right)$ samples from $\mathcal{D}$, one can output g that is ϵ-close to f with probability $\geq \frac{2}{3}$.

Proof: **High-degree weight bound:** By the total sensitivity formula,

$$\boldsymbol{S}_{\mathcal{D}}[f] = \sum_{k\geq 1} k \cdot W^k_{\mathcal{D}}[f] \geq \sum_{k>d} k \cdot W^k_{\mathcal{D}}[f] \geq d \cdot \sum_{k>d} W^k_{\mathcal{D}}[f]. \tag{35}$$

Thus $\sum_{k>d} W^k_{\mathcal{D}}[f] \leq \boldsymbol{S}_{\mathcal{D}}\frac{[f]}{d} \leq \frac{S}{d} \leq \frac{\epsilon^2}{4}$ when $d \geq 4\frac{S}{\epsilon^2}$.

Algorithm: For each $|\alpha| \leq d$, estimate $\hat{f}_{\mathcal{D}}(\alpha)$ by $\tilde{f}(\alpha) = \frac{1}{m}\sum_{j=1}^m f\left(x^{(j)}\right)\varphi_\alpha\left(x^{(j)}\right)$. Output $g = \sum_{|\alpha|\ \leq d} \tilde{f}(\alpha)\varphi_\alpha$.

Analysis: By Parseval for Dataset Closeness:

$$\mathbb{E}_{\mathcal{D}}[(f-g)^2] = \underbrace{\sum_{k>d} W^k_{\mathcal{D}}[f]}_{\leq \frac{\epsilon^2}{4}} + \underbrace{\sum_{|\alpha|\ \leq d} \left(\hat{f}_{\mathcal{D}}(\alpha) - \tilde{f}(\alpha)\right)^2}_{\leq \frac{\epsilon^2}{4} \text{ w.h.p.}}. \tag{36}$$

Total: $\mathbb{E}_{\mathcal{D}}\left[(f-g)^2\right] \leq \frac{\epsilon^2}{2} < \epsilon$. ■

3 Goldreich-Levin over Datasets

We now adapt the Goldreich-Levin (GL) algorithm ([1], Section 3.5) to the dataset setting. Throughout this section we work over the Boolean cube $\Omega^n = \{-1, 1\}^n$ with the parity characters $\chi_S(x) = \prod_{i\in S} x_i$ for clarity of exposition; we remark on general finite alphabets at the end.[2]

Recall the dataset Fourier coefficient

$$\hat{f}_{\mathcal{D}}(S) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x)] = \frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}} f(x)\chi_S(x),$$

and the normalization constant $C_{\mathcal{D}} = \frac{2^n}{|\mathcal{D}|}$. The algorithmic goal, mirroring the classical GL guarantee, is:

Definition 3.1 *(Dataset Heavy-Coefficient Search)*: Given $0 < \tau \leq 1$ and access to f and $\mathcal{D}$ (in a model to be specified), output a list L of subsets of $[n]$ such that with high probability:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$.

3.1 Access Models

The choice of access model turns out to be the crux of the problem. We distinguish three oracles:
1. **Samples** (SAMP): i.i.d. draws $x \sim \mathcal{D}$ together with the label $f(x)$.

[2] All identities below that only use orthonormality and the "reproducing kernel" property $\sum_S \chi_S(x)\chi_S(x') = 2^n \cdot \mathbb{1}[x = x']$ extend verbatim to any finite abelian group with its character basis (with $\chi_S\overline{\chi_T}$ in place of $\chi_S\chi_T$), and more generally to product orthonormal bases over finite sets.

2. **Membership** (MEM): the in-distribution tester $x \mapsto \mathcal{D}(x) \in \{0,1\}$, evaluable on *arbitrary* $x \in \{-1,1\}^n$.
3. **Model queries** (QUERY): f evaluable at arbitrary $x \in \{-1,1\}^n$, including out-of-distribution points. This is the natural model when f is a trained model or some other efficiently computable function.

Classical GL uses only QUERY, and its guarantee concerns the *global* (uniform-measure) coefficients $\hat{f}(S)$. We will see that for the *dataset* coefficients $\hat{f}_{\mathcal{D}}(S)$, samples alone are provably insufficient (Lemma 3.3), and that the positive results route the dataset structure through a single object: the dataset's *bias spectrum* (Lemma 3.4).

3.2 Recap: the Classical Goldreich-Levin Algorithm

Theorem 3.1 (Goldreich-Levin; [1] Section 3.5): Given query access to $f : \{-1,1\}^n \to \{-1,1\}$ and $0 < \tau \leq 1$, there is a $\mathrm{poly}(n, 1/\tau)$-time algorithm that with high probability outputs a list $L = \{U_1, ..., U_\ell\}$ of subsets of $[n]$ such that:
1. $|\hat{f}(U)| \geq \tau \Longrightarrow U \in L$;
2. $U \in L \Longrightarrow |\hat{f}(U)| \geq \tau/2$.

By Parseval's theorem, the second guarantee implies $|L| \leq 4/\tau^2$.

The engine of the algorithm is a divide-and-conquer search over the tree of coordinate prefixes. For $S \subseteq J \subseteq [n]$, define the **bucket weight** ([1], Definition 3.39)

$$W^{S \mid J}[f] = \sum_{T \subseteq \overline{J}} \hat{f}(S \cup T)^2,$$

the Fourier weight of f on sets whose restriction to J equals S. The algorithm maintains a collection of buckets of weight $\geq \tau^2/2$ (at most $4/\tau^2$ of them, by Parseval), splitting one coordinate at a time until each surviving bucket is a singleton $\{S\}$ with $\hat{f}(S)^2 \geq \tau^2/2$. Its feasibility rests on the restriction identity ([1], equation (3.5) and Proposition 3.40)

$$W^{S \mid J}[f] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\left[\hat{f}_{J \mid z}(S)^2\right] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\ \mathbb{E}_{y,y' \sim \{-1,1\}^J}[f(y,z)\chi_S(y) \cdot f(y',z)\chi_S(y')],$$

which lets a QUERY algorithm estimate any bucket weight to accuracy $\pm\epsilon$ with $O(\log(1/\delta)/\epsilon^2)$ queries.

Two remarks that we will use later. First, the proof only uses $\|f\|_\infty \leq 1$ (for the Chernoff estimates) and $\|f\|_2 \leq 1$ (for the Parseval pruning bound), so Theorem 3.1 holds verbatim for $f : \{-1,1\}^n \to [-1,1]$, with list size $4\|f\|_2^2/\tau^2$. Second, the running time is polynomial in $1/\tau$; this will be the quantity that blows up when we naively lift a sparse dataset.

3.3 Obstructions

Over a dataset, the characters $\{\chi_S\}$ are *not* orthonormal with respect to the measure $\mathrm{Unif}(\mathcal{D})$, and the familiar Fourier toolkit breaks in quantifiable ways. The first casualty is Parseval.

Lemma 3.1 *(Mass Identity)*: For any $\mathcal{D} \subseteq \{-1,1\}^n$ (distinct points) and any $f : \mathcal{D} \to \mathbb{R}$,

$$\sum_{S \subseteq [n]} \hat{f}_{\mathcal{D}}(S)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2].$$

Proof: Expanding the square and swapping sums,

$$\sum_S \hat{f}_{\mathcal{D}}(S)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x,x' \in \mathcal{D}} f(x) f(x') \sum_{S \subseteq [n]} \chi_S(x) \chi_S(x').$$

Since $\chi_S(x)\chi_S(x') = \chi_S(x \circ x')$ where $(x \circ x')_i = x_i x'_i$, the inner sum is $2^n \cdot \mathbb{1}[x = x']$ (the reproducing kernel of the character basis). Only the diagonal survives:

$$\sum_S \hat{f}_{\mathcal{D}}(S)^2 = \frac{2^n}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2].$$

■

Since $C_{\mathcal{D}}$ is exponentially large for small datasets, the total dataset Fourier mass is *not* bounded by $\mathbb{E}_{\mathcal{D}}[f^2]$: Parseval fails over $\mathcal{D}$ by exactly the factor $C_{\mathcal{D}}$. TODO: Lemma 3.1 contradicts the "Parseval for Dataset Closeness" lemma of the previous section, which asserts $\mathbb{E}_{x \sim \mathcal{D}}[(f-g)^2] = \sum_\alpha \left(\hat{f}_{\mathcal{D}}(\alpha) - \hat{g}_{\mathcal{D}}(\alpha)\right)^2$; the right-hand side is in fact $C_{\mathcal{D}}$ times the left. The basis $\{\varphi_\alpha\}$ is orthonormal in $L^2(\Omega^n, \mu)$, not in $L^2(\mathcal{D})$. The "Learning Low-Degree Fourier Functions" theorem built on that lemma needs to be revisited as well.

Two immediate consequences. First, the number of τ-heavy coefficients can be as large as $C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2]/\tau^2$, i.e., exponential; so no analogue of the list-size bound $|L| \leq 4/\tau^2$ can hold unconditionally. The subcube example shows this is not a pathology of weird functions — the constant function already exhibits it.

Example: Let $K \subseteq [n]$ and let $\mathcal{D} = \{x : x_i = 1 \text{ for all } i \in K\}$ be the subcube fixing K, with $f \equiv 1$. Then for every $S \subseteq K$ we have $\chi_S \equiv 1$ on $\mathcal{D}$, hence

$$\hat{f}_{\mathcal{D}}(S) = \begin{cases} 1 \text{ if } S \subseteq K \\ 0 \text{ otherwise,} \end{cases}$$

and there are $2^{|K|}$ maximal coefficients. On a dataset, distinct characters can be *identical* as functions on $\mathcal{D}$; "the" heavy set is only defined up to this aliasing.

Second — and more damning for the GL strategy — the bucket weights themselves carry *no information* over a generic dataset. The following exact identity is the dataset analogue of the restriction identity above.

Lemma 3.2 *(Pair Form of Dataset Bucket Weights)*: Fix $J \subseteq [n]$ with $|J| = k$ and $S \subseteq J$. Then

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = 2^{n-k} \cdot \mathbb{E}_{x,x' \sim \mathcal{D}}[f(x) f(x') \chi_S(x_J) \chi_S(x'_J) \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]].$$

Proof: Expand each $\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{x \sim \mathcal{D}}[f(x)\chi_S(x_J)\chi_U(x_{\overline{J}})]$, square, and swap sums:

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = \mathbb{E}_{x,x' \sim \mathcal{D}}\left[f(x) f(x') \chi_S(x_J) \chi_S(x'_J) \sum_{U \subseteq \overline{J}} \chi_U(x_{\overline{J}}) \chi_U(x'_{\overline{J}}) \right],$$

and the inner sum is the reproducing kernel $2^{n-k} \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]$ on $\{-1,1\}^{\overline{J}}$. ■

Lemma 3.3 *(Blindness of Dataset Bucket Weights)*: In the setting of Lemma 3.2, suppose the projection $x \mapsto x_{\overline{J}}$ is injective on $\mathcal{D}$ (no two dataset points collide on $\overline{J}$). Then for *every* $S \subseteq J$,

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = \frac{2^{n-k}}{|\mathcal{D}|} \cdot \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2],$$

independent of S.

Proof: Under injectivity, $\mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}] = \mathbb{1}[x = x']$ on $\mathcal{D} \times \mathcal{D}$, so only the diagonal of Lemma 3.2 survives, and $\chi_S(x_J)^2 = 1$. ■

In words: as long as the dataset's projection onto the un-split coordinates is collision-free, *every* bucket at level k has exactly the same weight, no matter what f is and no matter where its heavy dataset coefficients lie. For a dataset of m points in general position (e.g., m uniformly random points), a union bound gives collisions on $\overline{J}$ with probability at most $\binom{m}{2}/2^{n-k}$, so the search tree is information-free for all levels

$$k \lesssim n - 2\log_2 m :$$

the "birthday horizon." A GL-style search would have to descend blindly through $2^{n-2\log_2 m}$ buckets before bucket weights begin to distinguish anything. Together with the subcube example (which shows the target list itself can be exponential), this rules out a sample-only dataset GL: some additional access to f or structural knowledge of $\mathcal{D}$ is necessary.[3]

3.4 The Convolution Identity

The positive results all flow from one structural fact: the dataset spectrum of f is the global spectrum of f *convolved with the spectrum of the dataset itself.*

Definition 3.2 *(Bias Spectrum)*: The **bias spectrum** of $\mathcal{D}$ is the family

$$b_V = \mathbb{E}_{x\sim\mathcal{D}}[\chi_V(x)], \quad V \subseteq [n],$$

i.e., the dataset Fourier coefficients of the constant function 1. Note $b_\emptyset = 1$ and $|b_V| \leq 1$. For $\theta > 0$ we write $B_\theta = \{V : |b_V| \geq \theta\}$ for the **$\boldsymbol{\theta}$-heavy bias set**.

Lemma 3.4 *(Convolution Identity)*: For any $f : \{-1,1\}^n \to \mathbb{R}$ and any $S \subseteq [n]$,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V\subseteq[n]} b_V \cdot \hat{f}(S\Delta V),$$

where Δ denotes symmetric difference.

Proof: Expand f in its (exact, global) Fourier expansion $f = \sum_T \hat{f}(T)\chi_T$ inside the dataset expectation:

$$\hat{f}_{\mathcal{D}}(S) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x)] = \sum_T \hat{f}(T) \cdot \mathbb{E}_{x\sim\mathcal{D}}[\chi_T(x)\chi_S(x)] = \sum_T \hat{f}(T) \cdot b_{T\Delta S},$$

using $\chi_T\chi_S = \chi_{T\Delta S}$. Substituting $V = T\Delta S$ gives the claim. ■

The bias spectrum is exactly the aliasing structure of the subcube example: there, $b_V = \mathbb{1}[V \subseteq K]$, and the convolution smears the single global coefficient $\hat{f}(\emptyset) = 1$ onto all $2^{|K|}$ sets $S \subseteq K$. The blindness of Lemma 3.3 is the statement that, for a generic dataset, the bias spectrum is an exponentially wide field of $\sim \pm 1/\sqrt{|\mathcal{D}|}$ noise; the convolution then buries any bucket-level signal. Conversely, whenever the bias spectrum is *dominated by few heavy terms*, the dataset spectrum of f is a controlled deformation of its global spectrum — and that is precisely what the main theorem exploits.

[3] A sanity check for Lemma 3.3: take $n = 2$, $\mathcal{D} = \{(1,1),(-1,-1)\}$, $f(x) = x_1$, $J = \{1\}$. Then $\hat{f}_{\mathcal{D}}(\{1\}) = 1$, $\hat{f}_{\mathcal{D}}(\{1,2\}) = 0$, and the bucket weight is $1 = \frac{2^{2-1}}{2} \cdot 1$, matching the lemma; the "signal" coefficient and the aliasing noise conspire to a flat bucket profile. All identities in this section have additionally been verified numerically on random instances.

3.5 Main Theorem: Dataset GL from Model Queries

We now give the main positive result. The standing assumptions: f is a queryable model with bounded *spectral norm* $\|\hat{f}\|_1 = \sum_T |\hat{f}(T)|$, and we know the heavy part of the dataset's bias spectrum. The latter is a structural input about $\mathcal{D}$ alone (not about f); we discuss how to obtain it below.

Theorem 3.2 (Dataset Goldreich-Levin, model-query access)**:** Let $f : \{-1,1\}^n \to [-1,1]$ with $\|\hat{f}\|_1 \leq A$ be given by QUERY access, and let $\mathcal{D}$ be given by SAMP access. Let $0 < \tau \leq 1$, $\delta \in (0,1)$, fix any $\theta \leq \tau/(4A)$, and suppose the θ-heavy bias set B_θ of $\mathcal{D}$ is given explicitly. Then there is an algorithm running in time $\mathrm{poly}(n, |B_\theta|, 1/\tau, \log(1/\delta))$, using $\mathrm{poly}(n, |B_\theta|, 1/\tau) \cdot \log(1/\delta)$ queries to f and $O(\log(|B_\theta| / (\tau\delta))/\tau^2)$ samples from $\mathcal{D}$, that with probability $\geq 1 - \delta$ outputs a list L satisfying:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$;
3. $|L| \leq \frac{64\,|B_\theta|^3}{9\tau^2}$.

The algorithm:
1. Run classical GL (Theorem 3.1, in its $[-1,1]$-valued form) on f with threshold $\tau' = \frac{3\tau}{4\,|B_\theta|}$ and confidence $1 - \delta/2$, obtaining a list F of global heavy sets, $|F| \leq 4/\tau'^2$.
2. Form the candidate list $C = \{T \Delta V : T \in F, V \in B_\theta\}$, so $|C| \leq |F| \cdot |B_\theta|$.
3. Draw $m = O(\log(|C| / \delta)/\tau^2)$ samples from $\mathcal{D}$ and, for each $S \in C$, compute the empirical coefficient $\tilde{f}_{\mathcal{D}}(S) = \frac{1}{m}\sum_{j=1}^m f(x^{(j)})\chi_S(x^{(j)})$. Output $L = \{S \in C : |\tilde{f}_{\mathcal{D}}(S)| \geq 3\tau/4\}$.

Proof: **Decomposition.** By Lemma 3.4, for every S,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V \in B_\theta} b_V \hat{f}(S \Delta V) + R_S, \quad \text{where } |R_S| \leq \sum_{V \notin B_\theta} |b_V|\ |\hat{f}(S\Delta V)| \leq \theta \cdot \|\hat{f}\|_1 \leq \theta A \leq \frac{\tau}{4},$$

using $|b_V| < \theta$ off the heavy set and re-indexing the tail sum over $T = S \Delta V$.

Completeness. Suppose $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$. Then

$$\sum_{V \in B_\theta} |b_V|\ |\hat{f}(S\Delta V)| \geq |\hat{f}_{\mathcal{D}}(S)| - |R_S| \geq \tau - \frac{\tau}{4} = \frac{3\tau}{4},$$

and since $|b_V| \leq 1$, some $V^* \in B_\theta$ has $|\hat{f}(S\Delta V^*)| \geq \frac{3\tau}{4\,|B_\theta|} = \tau'$. By the completeness of classical GL (except with probability $\delta/2$), $S\Delta V^* \in F$, hence $S = (S\Delta V^*)\Delta V^* \in C$. By Hoeffding's inequality and a union bound over C (except with probability $\delta/2$), every empirical coefficient satisfies $|\tilde{f}_{\mathcal{D}}(S) - \hat{f}_{\mathcal{D}}(S)| \leq \tau/8$; in particular $|\tilde{f}_{\mathcal{D}}(S)| \geq \tau - \tau/8 > 3\tau/4$, so $S \in L$.

Soundness. If $S \in L$ then $|\hat{f}_{\mathcal{D}}(S)| \geq 3\tau/4 - \tau/8 = 5\tau/8 \geq \tau/2$.

List size and complexity. $|L| \leq |C| \leq |F|\ |B_\theta| \leq \frac{4}{\tau'^2}\ |B_\theta| = \frac{64\,|B_\theta|^3}{9\tau^2}$. Classical GL at threshold τ' costs $\mathrm{poly}(n, 1/\tau') = \mathrm{poly}(n, |B_\theta|, 1/\tau)$ queries; step 3 costs m samples and $m \cdot |C|$ arithmetic operations. ■

Note that the algorithm uses only the *set* B_θ, not the values b_V; and the theorem is meaningful precisely in the distribution-shift regime where $\hat{f}_{\mathcal{D}}(S)$ and $\hat{f}(S)$ differ — the shift is routed through the dataset's heavy biases. The same completeness argument, run purely combinatorially (Parseval bounds the number of τ'-heavy global coefficients by $1/\tau'^2$), yields an unconditional structural bound:

Corollary 3.1 *(List-Size Bound under Sparse Bias Spectrum)*: If $\left\|\hat{f}\right\|_1 \leq A$ and $\theta \leq \tau/(4A)$, then

$$|\left\{S : |\hat{f}_{\mathcal{D}}(S)| \geq \tau\right\}| \leq \frac{16\ |B_\theta|^3}{9\tau^2}.$$

3.6 Companion Results

Dense Datasets: the Membership-Oracle Route

When the dataset is *dense* — $C_{\mathcal{D}} = \mathrm{poly}(n)$ — the naive lifting strategy already works, and moreover supplies the heavy bias set B_θ needed above. Recall the lifted function $g = \mathcal{D} \circ f$, i.e., $g(x) = \mathcal{D}(x)f(x)$, which is evaluable anywhere given MEM together with the ability to evaluate f on points of $\mathcal{D}$; its global coefficients satisfy $\hat{g}(S) = C_{\mathcal{D}}^{-1}\hat{f}_{\mathcal{D}}(S)$.

Proposition 3.1 *(Dataset GL for Dense Datasets)*: Given query access to $g = \mathcal{D} \circ f$ with $\|f\|_\infty \leq 1$, and $0 < \tau \leq 1$, running classical GL on g with threshold $C_{\mathcal{D}}^{-1}\tau$ solves the search problem of Definition 3.1 in time $\mathrm{poly}(n, C_{\mathcal{D}}, 1/\tau)$, with list size $|L| \leq 4C_{\mathcal{D}}\mathbb{E}_{\mathcal{D}}[f^2]/\tau^2$.

Proof: g takes values in $[-1, 1]$, so the $[-1, 1]$-valued form of Theorem 3.1 applies at threshold $\tau'' = C_{\mathcal{D}}^{-1}\tau$: completeness and soundness transfer through $\hat{g}(S) = C_{\mathcal{D}}^{-1}\hat{f}_{\mathcal{D}}(S)$, the running time is $\mathrm{poly}(n, 1/\tau'') = \mathrm{poly}(n, C_{\mathcal{D}}, 1/\tau)$, and Parseval gives $|L| \leq 4\|g\|_2^2/\tau''^2 = 4C_{\mathcal{D}}^{-1}\mathbb{E}_{\mathcal{D}}[f^2] \cdot C_{\mathcal{D}}^2/\tau^2$. ■

The list-size bound matches the mass identity (Lemma 3.1) exactly, so Proposition 3.1 is tight in general; and by taking $f \equiv 1$, the same run recovers the heavy bias set: $b_V = C_{\mathcal{D}} \cdot \hat{\mathbb{1}}_{\mathcal{D}}(V)$, so B_θ is computable in time $\mathrm{poly}(n, C_{\mathcal{D}}, 1/\theta)$ given MEM. For sparse datasets ($C_{\mathcal{D}}$ superpolynomial) this route is unavailable — consistent with Lemma 3.3 — and B_θ must come from prior structural knowledge of the dataset (e.g., known feature constraints, class balance, or template structure).

Representative Datasets

Finally, if the dataset's spectrum uniformly tracks the global one — the "no distribution shift" regime — dataset GL degenerates gracefully to classical GL plus re-scoring.

Definition 3.3 *(ϵ-Representative Dataset)*: $\mathcal{D}$ is **ϵ-representative** for f if $|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)| \leq \epsilon$ for all $S \subseteq [n]$.

Lemma 3.5: Let $f : \{-1, 1\}^n \to [-1, 1]$ and let $\mathcal{D}$ consist of m i.i.d. uniform samples. If $m \geq \frac{2}{\epsilon^2}\left(n \ln 2 + \ln \frac{2}{\delta}\right)$, then with probability $\geq 1 - \delta$, $\mathcal{D}$ is ϵ-representative for f.

Proof: For fixed S, $\hat{f}_{\mathcal{D}}(S)$ is an average of m i.i.d. terms $f(x)\chi_S(x) \in [-1, 1]$ with mean $\hat{f}(S)$; Hoeffding gives $\Pr\left[\!\left[\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)| > \epsilon\right]\right. \leq 2e^{-m\epsilon^2/2}$. Union bound over all 2^n sets S. ■

Proposition 3.2 *(Dataset GL for Representative Datasets)*: Suppose $\mathcal{D}$ is $(\tau/8)$ -representative for $f : \{-1, 1\}^n \to [-1, 1]$, and we have QUERY access to f and SAMP access to $\mathcal{D}$. Then the search problem of Definition 3.1 is solvable in time $\mathrm{poly}(n, 1/\tau, \log(1/\delta))$ with $O(\log(1/(\tau\delta))/\tau^2)$ samples: run classical GL at threshold $7\tau/8$ to get F, estimate $\hat{f}_{\mathcal{D}}(S)$ for $S \in F$ to accuracy $\tau/8$ from samples, and keep $|\tilde{f}_{\mathcal{D}}(S)| \geq 3\tau/4$.

Proof: If $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$ then representativeness gives $|\hat{f}(S)| \geq \tau - \tau/8 = 7\tau/8$, so $S \in F$ by GL completeness; its estimate satisfies $|\tilde{f}_{\mathcal{D}}(S)| \geq |\hat{f}_{\mathcal{D}}(S)| - \tau/8 \geq 7\tau/8 > 3\tau/4$, so S is kept. (The representativeness slack is needed only for the GL step; the estimation step compares directly against $\hat{f}_{\mathcal{D}}(S)$.) If S is kept then $|\hat{f}_{\mathcal{D}}(S)| \geq 3\tau/4 - \tau/8 = 5\tau/8 \geq \tau/2$. GL's Parseval bound gives $|F| \leq 4/(7\tau/8)^2 \leq 7/\tau^2$, so a union bound over F needs $O(\log(|F|\,/\delta)/\tau^2)$ samples. ■

In this regime the heavy dataset coefficients essentially *are* the heavy global coefficients; the substantive new content of Theorem 3.2 lies exactly where representativeness fails.

3.7 Application: Kushilevitz-Mansour over Datasets

Classical GL powers the Kushilevitz-Mansour learning algorithm ([1], Theorems 3.37-3.38): any f with $\left\|\hat{f}\right\|_1 \leq s$ — in particular any decision tree of size s — is learnable from queries in time $\mathrm{poly}(n, s, 1/\epsilon)$. The spectral-norm hypothesis is exactly the A of Theorem 3.2, so we get the on-distribution analogue of the coefficient-collection step for free:

Corollary 3.2 *(Heavy On-Dataset Coefficients of Decision Trees)*: Let $f : \{-1,1\}^n \to \{-1,1\}$ be computable by a decision tree of size s (so $\left\|\hat{f}\right\|_1 \leq s$), given by QUERY access, and let B_θ be the θ-heavy bias set of $\mathcal{D}$ for some $\theta \leq \tau/(4s)$. Then all S with $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$ can be listed in time $\mathrm{poly}(n, s, |B_\theta|, 1/\tau, \log(1/\delta))$.

We emphasize what Corollary 3.2 does *not* yet give: an $\epsilon_{\mathcal{D}}$-close sparse approximation to f. Classically, coefficient collection plus Parseval yields the learning guarantee; over a dataset, Parseval fails (Lemma 3.1), so bounding $\mathbb{E}_{x\sim\mathcal{D}}[(f-g)^2]$ for a sparse g assembled from the recovered coefficients requires genuinely new arguments. TODO: On-dataset approximation from recovered coefficients. One route: $\mathbb{E}_{\mathcal{D}}[(f-g)^2] = \sum_V b_V \cdot \widehat{(f-g)^2}(V)$ by Lemma 3.4 applied to $(f-g)^2$ at $S = \emptyset$, splitting the sum over B_θ and its complement; the tail is controlled by $\theta \cdot \left\|\widehat{(f-g)^2}\right\|_1 \leq \theta\left(\left\|\hat{f}\right\|_1 + \|\hat{g}\|_1\right)^2$, but the heavy-bias terms need per-V control. Relatedly, fix the "Parseval for Dataset Closeness" lemma of the previous section.

3.8 Discussion and Open Directions

TODO:
- Non-uniform (weighted) datasets: all identities above should extend with $p(x)$ in place of $1/|\mathcal{D}|$; the reproducing-kernel steps are unaffected, the mass identity picks up $\|p\|_2^2$-type factors.
- General product alphabets and orthonormal bases (Hermite, tokens): the pair-form and blindness lemmas need only the reproducing kernel; the convolution identity needs the group structure ($\chi_S\overline{\chi_T} = \chi_{S-T}$), so state it over $\mathbb{Z}_q^n$ or general finite abelian groups.
- How to *certify* or *learn* the heavy bias set B_θ for structured real datasets (images, token corpora)? For dense datasets Proposition 3.1 computes it; for sparse ones, what natural structure (subcube constraints, sparsity templates, group symmetries) makes B_θ small and known?
- Relation to the "Active Fourier Auditor" of arXiv:2410.08111, which estimates distributional properties of ML models from queries and samples.
- Lower bound: turn Lemma 3.3 into a formal sample-complexity lower bound for Definition 3.1 in the SAMP-only model (two datasets, same visible statistics, different heavy sets).

4 Conclusion and Future Directions

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

Acknowledgments

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Bibliography

[1] Ryan O'Donnell. 2021. Analysis of boolean functions. *arXiv preprint arXiv:2105.10386* (2021).

APPENDIX A Appendix

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut.